

Jashinder Kumar | Electrical & Embedded Systems Engineering Student

Jammu, Jammu & Kashmir, India

+91 78899 87916 • jashinderkumar@gmail.com
in linkedin.com/in/yourprofile • github.com/jassi230202

Summary

Dedicated Electrical Engineering student specializing in embedded system hardware, power electronics conversion, and microcontrollers. Hands-on experience in designing power circuits, PCB layout, C/C++ firmware development, and practical substation operations. Seeking an engineering internship in embedded hardware, PCB design, or power electronics.

Education

MDU Rohtak <i>B.Tech in Electrical Engineering (2nd Year), 2nd Year</i> Coursework in Power Systems, Control Systems, Embedded Architectures, and Electrical Machines.	Rohtak, India <i>Present</i>
Govt Polytechnic College Jammu <i>Diploma in Electrical Engineering (3-Year Program), 3-Year Diploma</i> Full 3-year diploma curriculum in Electrical Machines, Circuit Analysis, Power Systems, and Electronics Laboratories.	Jammu, India <i>Completed</i>
Bal Bharti Public High School <i>Secondary School Certificate (10th)</i> Foundational education in Physics, Mathematics, Chemistry, and Computer Science.	Jammu, India <i>Completed</i>

Technical Skills

Hardware & CAD: Analog/Digital Circuit Design, KiCad EDA, Multisim, PCB Layout, Buck/Boost Converters

Embedded Firmware: C, C++, Arduino Framework, STM32 (Nucleo), ESP32 (Wi-Fi/Bluetooth), Python, MATLAB

Lab & Testing: Digital Oscilloscope, LTspice Simulation, Function Generator, Digital Multimeter

Power Electronics: Solar MPPT Controllers, PWM Motor Control, Smart BMS, Battery Storage Systems

Key Engineering Projects

Student Engineering Project <i>Lead Hardware Engineer, Solar Hybrid Buggy</i> Designed and built a solar-electric hybrid vehicle featuring roof-mounted PV panels, dual energy charging, PWM motor speed controller, and real-time battery voltage/current telemetry.	Jammu
Self-Initiated Project <i>Power Storage & BMS Lead, High-Capacity Battery Bank</i> Engineered a multi-cell battery pack with series-parallel topology. Integrated Smart BMS for active cell balancing, overcharge/thermal protection cutoffs, and digital state-of-charge (SOC) telemetry.	Jammu
Hands-on Project <i>Drivetrain Conversion, EV Bicycle Conversion</i> Converted a standard mechanical bicycle into an electric bicycle utilizing a Brushless DC (BLDC) Hub Motor, electronic twist throttle, lithium battery pack, and E-ABS brake power cutoff safety levers.	Jammu

Experience & Training

Gladni Grid Station (220kV / 132kV)

Jammu, India

Grid Station Operations

4-Week Internship

Gained practical field exposure to high-voltage switchgear, SF6 circuit breakers, power transformers, air-break disconnectors, busbar protection schemes, and numerical relay testing.

Certified Renewable Energy Training

Jammu, India

Solar Power Technologies

4-Week Certification

Trained in solar PV system design, battery storage sizing, PWM vs. MPPT charge controller selection, and grid-tied/off-grid system integration.